

Karthikeyan Devadoss

Forward Deployed Engineer (search & agent infrastructure) | RAG & backend foundation

Berlin, Germany · German permanent residence (Niederlassungserlaubnis) · no visa sponsorship needed · Live AI platform: [agentic-platform-backend-production.up.railway.app](#) · [github.com/karthikdevadoss](#)

SUMMARY

Applying for Forward Deployed Engineer, Enterprise at Tavily. Engineer focused on AI solution architecture, with about ten years of backend development since December 2016. Since early 2026, one of a handful of engineers in an initial core AI team at a large retail-energy company, designing and building AI solutions hands-on alongside senior backend work, including production AI triage with a human approval step, adopted by several teams. 4+ years combining hands-on Java/Spring with architecture discussions and design decisions alongside the lead.

CORE SKILLS

AI stack	Models used day-to-day: Claude Sonnet, Claude Code, Grok Bot, Fabel, GPT-4o / Azure OpenAI, Gemini · vector retrieval (production triage) · Python, Anthropic API, embeddings, Postgres, Playwright (personal project)
AI architecture	AI solution design · human-approval & capability boundaries · deterministic vs LLM decision split · RAG & retrieval evaluation (recall/MRR) · MCP & agent tool boundaries
Backend	Java, Spring Boot, REST, GraphQL, JWT/FusionAuth, AWS Lambda/SAM, SQS, Oracle, Docker, Jenkins, JUnit, Kafka, FHIR, Apache Camel

EXPERIENCE

Senior Backend Developer — NRG Energy

Mar 2022 – present

End client; employed via Remote GmbH · Texas, USA / Berlin

- Java/Spring Boot on a large retail-energy platform serving 5+ million customers; since early 2026, one of a handful of engineers in the initial core AI team — designing and building AI solutions hands-on (production AI triage with Azure OpenAI, vector retrieval, and human email approval before action; adopted by several teams) alongside senior backend work.
- Took part in architecture discussions and design decisions with the lead; REST, GraphQL, JWT/FusionAuth; 2025 AWS Lambda/SAM; mentoring; monthly production on-call.

Java Developer Consultant — Blue Cross Blue Shield Association (BCBSA)

Jun 2019 – Mar 2022

Chicago, USA

- National hub connecting state Blue Cross Blue Shield plans (~118 million members system-wide): Java/Spring Boot REST and FHIR interoperability / payer-to-payer exchange for member, coverage, and claims services.
- Application-side Kafka producers and consumers for domain events; careful handling of sensitive healthcare data.

Java Backend Developer — Marsh (Digital Broker / Bluestream, Affinity)

Dec 2016 – Jun 2019

Via Innovecture LLC · Phoenix, USA · ~95,000 colleagues, 130 countries

- Java/Spring carrier integrations (Apache Camel routes, ACORD-oriented mapping); on IMT-Germany, Data Modeler and System Analyst — built within the technical architect's design.

PERSONAL PROJECT

Agentic Software Delivery Platform

PERSONAL PROJECT

Live demo · GitHub repo

Independently designed and built

- Deterministic vs LLM split:** deterministic checks kept separate from LLM proposals; a Python agent (Anthropic API) plans and applies changes to a real Spring Boot target via tool calling.
- Human-approval boundary:** structural approval gate; the model has no code path to approve its own change, and approval is bound by content hash to the exact change.
- RAG eval design:** local retrieval (embeddings, cosine similarity, content-hash indexing) with a labelled eval (recall/MRR).
- MCP / tool boundaries:** MCP server for read-only repository tools; independent QA subagent without write tools; durable Postgres event ledger with versioned pricing.

EDUCATION

B.E. Computer Science — Madras University, Chennai

LANGUAGES

English — fluent · German — A1-A2